

FTEC5660 Assignment 2 part 1

Name: LI Shaoxuan

1. System Architecture and Design Decisions

Architecture Overview: The automated CV verification system is built upon a decoupled architecture to maximize stability, accuracy, and token efficiency. It consists of two primary modules:

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- **Information Extraction Module:** Instead of feeding raw text directly into the agent's reasoning loop, the system first employs the **Gemini 2.5 Flash** model (integrated via Google Cloud Vertex AI) utilizing its `with_structured_output` feature constrained by Pydantic schemas. This step cleanly extracts structured entities (Name, Current Company, Education, Locations) into a JSON format, providing precise parameters for subsequent tool usage and preventing the LLM from losing context in lengthy documents.
- **Agentic Verification Module (LangGraph):** The core reasoning engine utilizes state-of-the-art **LangGraph** (`create_react_agent`) instead of legacy agent executors. Operating in a ReAct (Reasoning and Acting) loop, the agent connects to the provided Model Context Protocol (MCP) server. It iteratively searches, retrieves, and cross-references data against the extracted CV JSON.

Design Decisions & Robustness:

- **Asynchronous Tool Execution:** Because MCP server tools inherently involve network requests for social media verification, the system is designed to trigger the agent using asynchronous invocation (`ainvoke`). This resolves blocking issues and strictly aligns with LangChain's asynchronous tool requirements.
- **Robust Output Parsing:** To reliably extract the final verification score for the evaluation threshold (0.5), the Agent is instructed to output its final reasoning and score in a valid JSON format. To handle the list-based output structures unique to certain LLM versions, the system utilizes a cascading parsing strategy: it first attempts Regex extraction (`re.search`), followed by fallback substring matching (locating `{` and `}`), ensuring the float score is always safely extracted.
- **Error Handling:** The execution loops are wrapped in try-except blocks. If an

extraction fails or the agent encounters a parsing error, it safely assigns a default score of 0.0, ensuring the batch evaluation process remains uninterrupted.

- **Edge Case Handling via Prompt Engineering:** During testing, I discovered that the MCP LinkedIn API occasionally returns "is_current": false alongside "end_year": null for active roles. To prevent the LLM from falsely penalizing valid CVs (like CV_1 and CV_3) due to this API quirk, a specific critical rule was added to the ReAct system prompt to force a positive evaluation logic in this scenario.

2. Agent Workflow and Tool Usage Strategy

The agent is governed by a strict System Prompt to enforce a deterministic, step-by-step strategy:

1. **Primary Search (LinkedIn):** The workflow strictly begins with the search_linkedin_people tool. The agent inputs the candidate's structured name and location.
2. **Profile Retrieval:** Using the ID obtained from the search, the agent calls get_linkedin_profile to retrieve the single source of truth for professional history.
3. **Cross-Checking Logic:** The agent critically compares the extracted CV data against the retrieved LinkedIn profile. It specifically evaluates matching employment timelines, job titles, and educational degrees.
4. **Fallback Strategy (Facebook):** If LinkedIn data is insufficient, the agent is instructed to use search_facebook_users and get_facebook_profile for secondary verification.
5. **Scoring Strategy:**
 - **Score > 0.5 (Valid):** Awarded when CV timelines and companies generally align with the social graph data. Minor formatting differences are tolerated.
 - **Score <= 0.5 (Invalid):** Awarded immediately upon detecting severe discrepancies, such as fabricated degrees, conflicting dates, or non-existent companies.

3. Sample Verification Results

The implemented system successfully evaluated the 5 sample CVs, achieving a perfect evaluation score of 1.0 (5/5 correct). The threshold used for binary classification was 0.5. The

incorporation of the CRITICAL RULE in the prompt effectively handled API edge cases, resulting in stable and highly accurate reasoning.

- **CV_1 (John Smith) - Score: 0.9 (Valid)**
 - **Reasoning:** All key details, including the role at ByteDance (Engineer, 2020) and education at McGill University, matched the retrieved profile perfectly. The agent correctly applied the prompt's critical rule to interpret "is_current": false alongside "end_year": null as a valid "Present" employment status, preventing a false penalty.
- **CV_2 (Minh Pham) - Score: 1.0 (Valid)**
 - **Reasoning:** Cross-referencing confirmed a perfect match across all fields, including the current role at BCG (Manager, 2022-Present), past experience at Tencent (2013-2017), and the BSc from The University of Hong Kong. No discrepancies were found.
- **CV_3 (Wei Zhang) - Score: 0.95 (Valid)**
 - **Reasoning:** The consulting history at PwC (Engineer, 2013) and degree from the University of Tokyo were successfully verified. Similar to CV_1, the system intelligently handled the end_year: null API quirk to validate the candidate's "Present" employment claim, resulting in a highly confident score.
- **CV_4 (Rahul Sharma) - Score: 0.1 (Invalid)**
 - **Reasoning:** The agent detected critical discrepancies. While the CV claimed a Senior Engineer role at Microsoft (with a highly suspicious date of 2021-2027) and a PhD from Tsinghua, the verified LinkedIn profile indicated an Analyst role at Tencent and an MSc from Stanford. The CV was correctly flagged as fabricated.
- **CV_5 (Rahul Sharma) - Score: 0.1 (Invalid)**
 - **Reasoning:** The agent demonstrated advanced multi-step reasoning by exhaustively searching multiple LinkedIn and Facebook profiles. It verified that none of the claimed experiences (EY, StartupXYZ, DataForge) or education (PhD at Univ. of Tokyo) matched any of the retrieved digital footprints for individuals with this name and location. The CV was decisively invalidated.